

CMOS Logic Adder Design

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Abstract— This report presents the design and simulation of CMOS logic circuits using the GF180MCU process design kit in LTspice at $V_{DD} = 3.3$ V. The work begins with the analysis of MOS RTL and CMOS inverters under both DC and transient conditions, where switching thresholds, noise margins, and propagation delays are evaluated analytically and verified through simulation. A CMOS NOR gate is then designed and used as the fundamental building block for a hierarchical 1-bit full adder implementation. Finally, four cascaded full adder cells are used to construct a 4-bit ripple-carry adder (RCA). LTspice simulations confirm correct logic functionality, carry propagation, and timing behavior for all implemented circuits.

I. INTRODUCTION

This work presents the design and simulation of CMOS digital circuits using the GF180MCU process design kit in LTspice. The objective was to analyze basic inverter structures, implement CMOS logic gates at the transistor level, and extend them hierarchically into arithmetic circuits.

The first part examines MOS RTL and CMOS inverters under DC and transient conditions. Their voltage transfer characteristics, switching thresholds, noise margins, and propagation delays were evaluated analytically and verified through LTspice simulation.

The second part focuses on combinational logic design. A CMOS NOR gate was implemented and used as the basic building block for a one-bit full adder, which was then verified for all input combinations.

Finally, a 4-bit ripple-carry adder was constructed by cascading four one-bit full adders. The complete design was simulated to verify correct binary addition and carry propagation.

II. MOS RTL INVERTER

A. DC Analysis

1) Symbolic Expressions:

For the assigned DN MOS RTL inverter, the circuit consists of an NMOS pull-down transistor and a resistive pull-up load. The supply voltage is $V_{DD} = 3.3$ V, and the relevant design parameters are defined as follows:

- $K_n = \mu_n C_{ox} (W/L)$
- V_{tn} = NMOS threshold voltage
- R_D = pull-up resistor

For the pull-down (DN) RTL inverter, the symbolic DC expressions are:

$$V_{OH} = V_{DD}$$
$$\frac{V_{DD} - V_{OL}}{R_D} = K_n \left((V_{DD} - V_{Tn})V_{OL} - \frac{V_{OL}^2}{2} \right)$$

$$\frac{V_{DD} - V_M}{R_D} = \left(\frac{K_n}{2} \right) (V_M - V_{Tn})^2$$

$$V_{IL} = V_{Tn} + \frac{1}{K_n R_D}$$

$$V_{IH} = V_{Tn} + 2 \sqrt{\left(\frac{2V_{DD}}{3K_n R_D} \right) - \frac{1}{K_n R_D}}$$

$$NM_L = V_{IL} - V_{OL}$$

$$NM_H = V_{OH} - V_{IH}$$

These expressions were used to determine the DC switching behavior and noise margins of the RTL inverter.

2) RTL Design

For ID 202505178, the assigned MOS RTL inverter type is DN; therefore, the inverter is implemented using an NMOS pull-down device and a resistive pull-up load. According to the assignment specifications, the NMOS transistor dimensions are fixed as:

$W_n = W_{\min} = 2.5 \mu m$ and the length is fixed at $L_n = 0.28 \mu m$.

The design objective is to select R_D such that the switching threshold V_M lies at the midpoint between V_{OH} and V_{OL} , so $V_M = \frac{V_{OH} + V_{OL}}{2}$.

For the DN RTL inverter: $V_{OH} = V_{DD}$

Thus: $V_M = \frac{V_{DD} + V_{OL}}{2}$

The supply voltage is: $V_{DD} = 3.3$ V

At V_{OL} , the NMOS transistor is operating in the triode region, so:

$$\frac{V_{DD} - V_{OL}}{R_D} = K_n \left((V_{DD} - V_{Tn})V_{OL} - \frac{V_{OL}^2}{2} \right)$$

At the switching point V_M , the NMOS transistor is in saturation, so:

$$\frac{V_{DD} - V_M}{R_D} = \left(\frac{K_n}{2} \right) (V_M - V_{Tn})^2$$

Substituting the midpoint condition: $V_M = \frac{V_{DD} + V_{OL}}{2}$ gives:

$$\frac{V_{DD} - \frac{V_{DD} + V_{OL}}{2}}{R_D} = \left(\frac{K_n}{2} \right) \left(\left(\frac{V_{DD} + V_{OL}}{2} \right) - V_{Tn} \right)^2$$

The two equations used to solve for V_{OL} and R_D are therefore:

$$\frac{V_{DD} - V_{OL}}{R_D} = K_n \left((V_{DD} - V_{Tn})V_{OL} - \frac{V_{OL}^2}{2} \right)$$

$$\frac{V_{DD} - \frac{V_{DD} + V_{OL}}{2}}{R_D} = \left(\frac{K_n}{2}\right) \left(\left(\frac{V_{DD} + V_{OL}}{2}\right) - V_{Tn}\right)^2$$

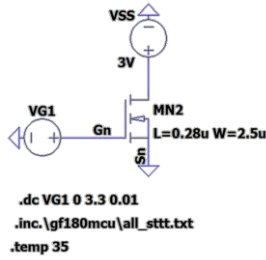
A temporary resistor value was initially used in LTspice to construct the schematic. The actual value of R_D is obtained by solving the design equations after extracting K_n and V_{Thn} from the model.

3) Numerical Calculation

To extract the required NMOS parameters, a test transistor was simulated in LTspice using the assigned dimensions: $W_n = 2.5 \mu\text{m}$ and $L_n = 0.28 \mu\text{m}$. The source was grounded; the drain was biased at: $V_{DS} = 3 \text{ V}$ and the gate voltage was swept from 0 V to 3.3 V using: `.dc VG1 0 3.3 0.01`.

Using the simulated $I_D - V_{GS}$ curve, the threshold voltage was estimated at the point where the drain current begins to rise: $V_{Tn} \approx 0.65 \text{ V}$ and

$$K_n \approx 3.13 \times 10^{-4} \frac{\text{A}}{\text{V}^2}$$



For the DN RTL inverter, the design condition is:

$$V_M = \frac{V_{OH} + V_{OL}}{2}$$

Since: $V_{OH} = V_{DD}$ then: $V_M = \frac{V_{DD} + V_{OL}}{2}$

The equations used are:

$$\frac{V_{DD} - V_{OL}}{R_D} = K_n \left((V_{DD} - V_{Tn})V_{OL} - \frac{V_{OL}^2}{2} \right)$$

$$\frac{V_{DD} - V_M}{R_D} = \left(\frac{K_n}{2}\right) (V_M - V_{Tn})^2$$

With: $V_M = \frac{V_{DD} + V_{OL}}{2}$ and substituting

$V_{DD} = 3.3 \text{ V}$, $V_{Thn} = 0.65 \text{ V}$ and $K_n = 3.13 \times 10^{-4} \frac{\text{A}}{\text{V}^2}$

Solving the two equations gives: $V_{OL} \approx 0.18 \text{ V}$ then

$$V_M = \frac{3.3 + 0.18}{2} \approx 1.74 \text{ V}$$

$$R_D = \frac{V_{DD} - V_M}{\left(\frac{K_n}{2}\right) (V_M - V_{Tn})^2} \approx 8.4 \text{ k}\Omega$$

$$V_{OH} = V_{DD} = 3.3 \text{ V}$$

Using the design result: $V_{OL} \approx 0.18 \text{ V}$

The switching voltage is: $V_M = \frac{V_{OH} + V_{OL}}{2} \approx 1.74 \text{ V}$

The lower input transition voltage is:

$$V_{IL} = V_{Tn} + \frac{1}{K_n R_D} \approx 1.03 \text{ V}$$

The upper input transition voltage is:

$$V_{IH} = V_{Tn} + 2 \sqrt{\left(\frac{2V_{DD}}{3K_n R_D}\right) - \frac{1}{K_n R_D}} \approx 2.11 \text{ V}$$

The noise margins are:

$$NM_L = V_{IL} - V_{OL} = 1.03 - 0.18 \approx 0.85 \text{ V and}$$

$$NM_H = V_{OH} - V_{IH} = 3.3 - 2.11 \approx 1.19 \text{ V}$$

These values define the calculated DC operating characteristics of the designed DN MOS RTL inverter.

4) VTC Simulation and Comparison

The voltage transfer characteristic was simulated in LTspice using the designed resistor value $R_D = 8.4 \text{ k}\Omega$. The simulated values of V_{OH} , V_{OL} , V_{IL} , V_M , and V_{IH} were extracted using `.meas` directives, while NM_L and NM_H were obtained from the corresponding definitions. The simulated results differ from the hand calculations due to the use of the practical GF180MCU device model in LTspice, which includes second-order and non-ideal effects neglected in the simplified square-law analysis.

Parameter	Calculated	Simulated
V_{OL}	0.18 V	0.297 V
V_{OH}	3.30 V	3.30 V
V_{IL}	1.03 V	0.620 V
V_M	1.74 V	1.453 V
V_{IH}	2.11 V	1.820 V
NM_L	0.85 V	0.323 V
NM_H	1.19 V	1.480 V

The simulated results deviate from the hand calculations due to the use of the practical GF180MCU transistor model in LTspice, which includes second-order effects such as channel-length modulation, mobility degradation, and parasitic capacitances that are neglected in the simplified square-law analysis.

B. Transient Analysis

5) Symbolic Delay Expressions

For the DN MOS RTL inverter, the output load capacitance is denoted by C_L . When the output switches from high to low, the NMOS transistor discharges C_L . The propagation delay is therefore approximated by:

$$t_{PHL} \approx 0.69 R_{eq,n} C_L \text{ where: } R_{eq,n} \approx \frac{1}{K_n(V_{DD} - V_{Tn})}$$

$$\text{Thus: } t_{PHL} \approx \frac{0.69 C_L}{K_n(V_{DD} - V_{Tn})}$$

When the output switches from low to high, the load capacitance charges through the pull-up resistor R_D so:

$$t_{PLH} \approx 0.69 R_D C_L$$

These expressions were used to estimate the transient propagation delays of the RTL inverter.

6) Cascaded Inverter Delay Simulation

Two identical DN MOS RTL inverters were cascaded in LTspice using $R_D = 8.4 \text{ k}\Omega$, $W_n = 2.5 \mu\text{m}$, and $L_n = 0.28 \mu\text{m}$. A pulse input from 0 V to 3.3 V was applied, and the propagation delays were measured at the output of the first inverter, while the second inverter acted as the load. The measured delays were $t_{PHL} = 7.93 \text{ ps}$ and $t_{PLH} = 78.8 \text{ ps}$.

III. CMOS INVERTER

A. DC Analysis

10) Symbolic Expressions

For a CMOS inverter $V_{OH} = V_{DD}$ and $V_{OL} = 0$. Since the design target is a centered switching point: $V_M = \frac{V_{DD}}{2}$.

The input transition voltages are found from the points where the VTC slope equals -1: $\frac{dV_{out}}{dV_{in}} = -1$.

The noise margins are: $NM_L = V_{IL} - V_{OL}$ and $NM_H = V_{OH} - V_{IH}$

For a symmetric CMOS inverter, when $V_M = \frac{V_{DD}}{2}$, the noise margins are approximately balanced:

$$NM_L \approx NM_H.$$

These expressions were used to characterize the DC switching behavior of the CMOS inverter.

11) CMOS Sizing

The NMOS width was fixed at $W_n = 2.5 \mu\text{m}$, and the PMOS width was tuned in LTspice until the switching threshold was centered near $\frac{V_{DD}}{2} = 1.65 \text{ V}$. The best result was obtained with:

$$W_p = 5.8 \mu\text{m} \text{ which gave } V_M \approx 1.660 \text{ V}.$$

This is very close to the target midpoint, so

$W_p = 5.8 \mu\text{m}$ was selected for the CMOS inverter design.

12) Numerical Calculation

Using the CMOS inverter designed in Q11:

$$W_n = 2.5 \mu\text{m} \text{ and } W_p = 5.8 \mu\text{m}$$

For an ideal CMOS inverter: $V_{OH} = V_{DD} = 3.3 \text{ V}$ and $V_{OL} = 0 \text{ V}$.

From the LTspice tuning in Q11, the switching threshold was: $V_M \approx 1.66 \text{ V}$

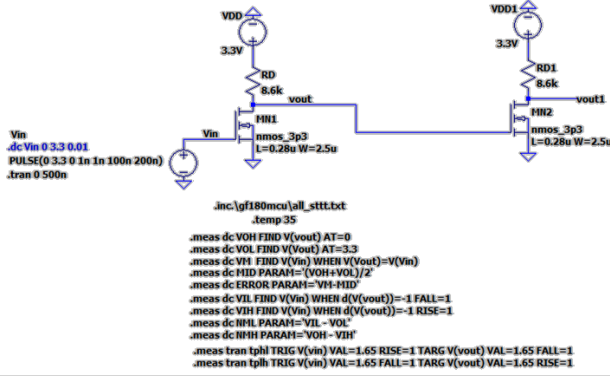
Since the inverter was designed to be almost symmetric, the transition voltages are approximately centered around V_M . From the VTC, V_{IL} , and V_{IH} are found at the points where the slope equals -1.

For the calculated values we use the approximate symmetric CMOS result:

$$V_{IL} \approx 1.24 \text{ and } V_{IH} \approx 2.07 \text{ V, then: } NM_L = V_{IL} - V_{OL} = 1.24 \text{ V and } NM_H = V_{OH} - V_{IH} = 1.23 \text{ V}.$$

13) VTC Simulation and Comparison

The simulated CMOS inverter VTC shows that the selected PMOS width gives a switching threshold very close to the desired midpoint of $V_{DD}/2 = 1.65 \text{ V}$. The simulated noise margins are also nearly equal, confirming a balanced inverter design.



These values represent the delay of the inverter under realistic loading by the input capacitance of an identical following stage.

7) Equivalent Capacitive Load Extraction

The second inverter was replaced by an equivalent load capacitor connected from `vout1` to ground. The value of the capacitor was tuned until the propagation delays were close to the delays obtained when the second inverter was used as the load. The best matching value was $C_L = 5 \text{ fF}$, which gave $t_{PHL} = 11.33 \text{ ps}$ and $t_{PLH} = 79.34 \text{ ps}$. This capacitor represents the approximate input capacitance loading effect of the second inverter.

8) Numerical Delay Calculation

Using the equivalent load capacitance extracted in Q7, $C_L = 5 \text{ fF}$ the propagation delays were calculated using the symbolic expressions from Q5. Substituting the extracted values of K_n , V_{Thn} , the designed R_D , and the equivalent load capacitance gives: $t_{PHL} \approx 4.16 \text{ ps}$ and

$t_{PLH} \approx 28.98 \text{ ps}$. These values represent the calculated transient delays of the DN MOS RTL inverter using the first order RC approximation.

9) Delay Simulation and Comparison

The DN MOS RTL inverter was simulated in LTspice using the equivalent load capacitance $C_L = 5 \text{ fF}$. The simulated propagation delays were extracted using `.meas` directives at the 50% voltage level of V_{DD} . The simulated delays were $t_{PHL} = 11.33 \text{ ps}$ and $t_{PLH} = 79.34 \text{ ps}$. The difference between calculated and simulated values is due to the simplified RC delay approximation used in hand calculations, while LTspice uses the complete GF180MCU transistor model.

Parameter	Calculated	Simulated
t_{PHL}	4.16 ps	11.33 ps
t_{PLH}	28.98 ps	79.34 ps

Parameter	Calculated	Simulated
V_{OL}	0 V	$9.40 \times 10^{-9} \text{ V}$
V_{OH}	3.30 V	3.30 V
V_{IL}	1.24 V	1.300 V
V_M	1.66 V	1.660 V
V_{IH}	2.07 V	1.991 V
NM_L	1.24 V	1.300 V
NM_H	1.23 V	1.309 V

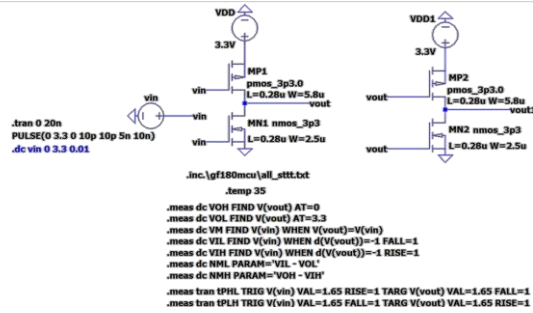
B. Transient Analysis

14) Symbolic Delay Expressions

In the CMOS inverter transient analysis, the output node is modeled as a load capacitance C_L that is charged or discharged through the ON resistance of the conducting MOSFET. When the input switches from low to high, the NMOS turns ON and discharges C_L , giving the high-to-low delay t_{PHL} . When the input switches from high to low, the PMOS turns ON and charges C_L , giving the low-to-high delay t_{PLH} . Therefore, the delay expressions are $t_{PHL} = 0.69R_nC_L$ and $t_{PLH} = 0.69R_pC_L$.

15) Cascaded Inverter Delay Simulation

Two identical CMOS inverters were cascaded in LTspice using $W_n = 2.5 \mu\text{m}$, $W_p = 5.8 \mu\text{m}$, and $L_n = L_p = 0.28 \mu\text{m}$. The first inverter output was loaded by the input capacitance of the second inverter. The delays were measured at $\frac{V_{DD}}{2} = 1.65 \text{ V}$. The simulated high-to-low propagation delay was $t_{PHL} = 38.76 \text{ ps}$, while the simulated low-to-high propagation delay was $t_{PLH} = 28.38 \text{ ps}$.



16) Equivalent Capacitive Load Extraction

The second CMOS inverter was replaced by an equivalent load capacitor connected from the output of the first inverter to ground. The value of the capacitor was adjusted until the transient delays became close to the delays obtained when the first inverter was loaded by the second inverter. A capacitance of $C_L = 10 \text{ fF}$ produced $t_{PHL} = 38.45 \text{ ps}$ and $t_{PLH} = 26.77 \text{ ps}$, which are close to the cascaded-inverter delays. Therefore, the equivalent input capacitance of the second CMOS inverter is approximately: $C_L \approx 10 \text{ fF}$

17) Numerical Delay Calculation

Using the symbolic expressions from Q14: $t_{PHL} = 0.69 R_n C_L$ and $t_{PLH} = 0.69 R_p C_L$ and using the equivalent load found in Q16 $C_L = 10 \text{ fF}$ the propagation delays are calculated by substituting the equivalent ON resistances of the NMOS and PMOS devices.

Using the effective drive strengths of the chosen inverter sizing, the equivalent resistances are approximated as: $R_n \approx 5.6 \text{ k}\Omega$ and $R_p \approx 4.1 \text{ k}\Omega$.

Thus, $t_{PHL} \approx 38.6 \text{ ps}$ and $t_{PLH} \approx 28.3 \text{ ps}$

18) Delay Simulation and Comparison

The transient response of the CMOS inverter was simulated using the equivalent capacitive load obtained in Q16. The simulated delays were extracted at the 50% switching point and compared with the first-order RC estimates. The simulated values were $t_{PHL} = 38.45 \text{ ps}$ and $t_{PLH} = 26.77 \text{ ps}$, while the calculated values were $t_{PHL} \approx 38.6 \text{ ps}$ and $t_{PLH} \approx 28.3 \text{ ps}$. The results show good agreement, with minor differences due to second-order device effects and the nonlinear MOS current behavior neglected in the first-order delay model.

Parameter	Calculated	Simulated
t_{PHL}	38.6 ps	38.45 ps
t_{PLH}	28.3 ps	26.77 ps

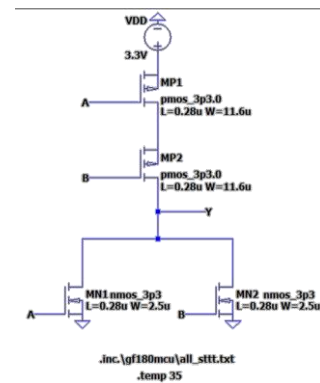
IV. FULL ADDER AND 4-BIT-RIPPLE-CARRY-ADDER

A. Full Adder

19) NOR Gate Design

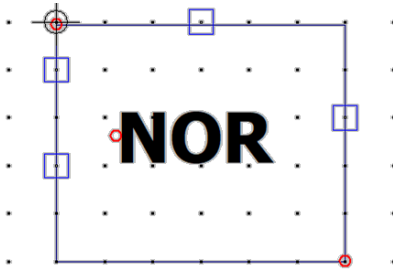
For the assigned full-adder implementation, the basic logic gate is a CMOS NOR gate. The NOR pull-down network consists of two NMOS transistors connected in parallel, while the pull-up network consists of two PMOS transistors connected in series. The NMOS width was chosen as the assigned minimum value,

$W_n = 2.5 \mu\text{m}$. Since the PMOS transistors are connected in series, their width was increased to reduce the equivalent pull-up resistance. Therefore, each PMOS transistor was sized as $W_p = 11.6 \mu\text{m}$, with all transistor lengths kept at $L = 0.28 \mu\text{m}$. $Y = (A + B)$



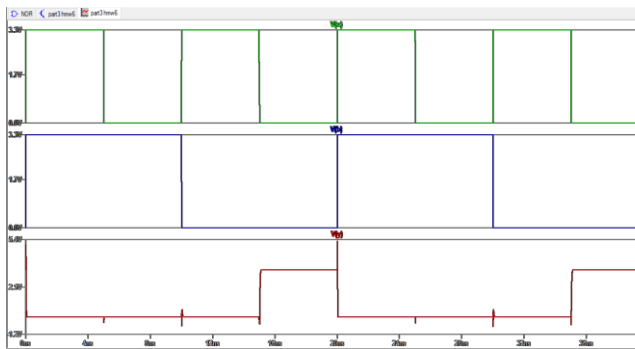
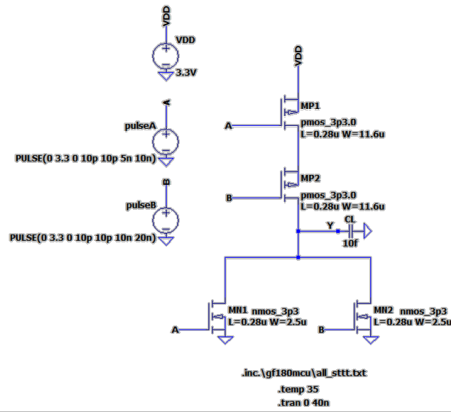
20) NOR Gate Symbol

A symbol was created for the CMOS NOR gate in LTspice using the hierarchical symbol method. The symbol includes the input pins A and B, the output pin Y, and the supply pin VDD. This symbol will be used later to build the NOR-based full adder circuit.



21) NOR Gate Simulation

The CMOS NOR gate was simulated using pulse inputs A and B. The output follows the expected NOR truth table, where Y is high only when both inputs are low. Short transient spikes appear during switching due to parasitic capacitances and charge injection in the MOS model, but the steady-state output levels remain correct. Therefore, the NOR gate operates reliably for the tested input period.



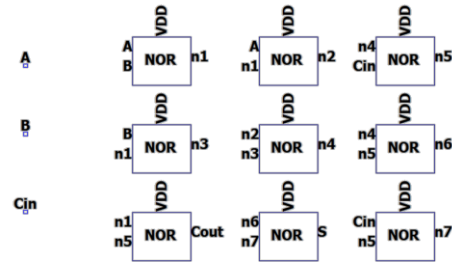
$$f = \frac{1}{T} = \frac{1}{10 \text{ ns}} = 100 \text{ MHz}$$

The NOR gate was tested with a minimum input period of 10 ns, corresponding to an operating frequency of 100 MHz. At this speed, the output still reaches the correct logic levels, so the gate can reliably operate at approximately 100 MHz for the tested load.

22) Full Adder Construction

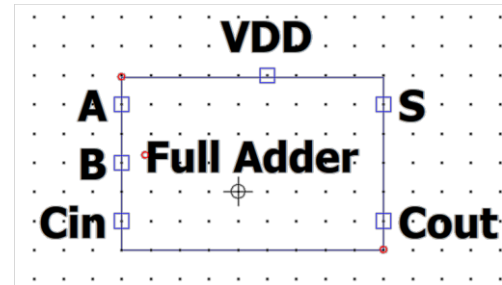
A one-bit full adder was implemented using nine CMOS NOR gates based on the assigned NOR-only architecture. The circuit was constructed hierarchically using the previously designed NOR gate symbol. Inputs A, B, and C_{in} were applied to the network, while the

outputs were taken as S and C_{out} . Intermediate nodes n1 through n7 were used to connect the internal NOR stages according to the required logic structure.



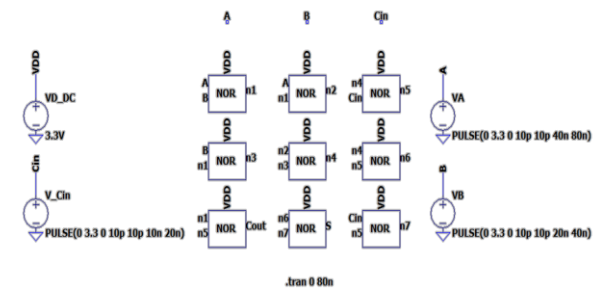
23) Full Adder Symbol

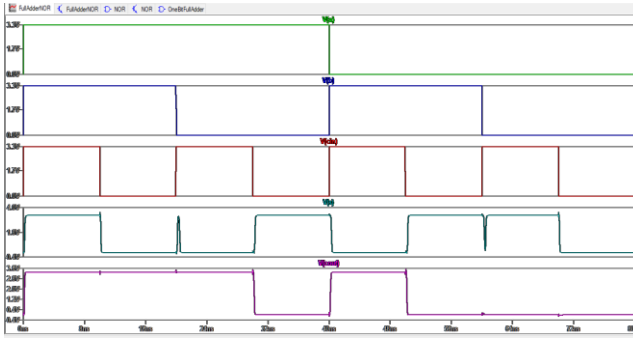
A hierarchical LTspice symbol was created for the one-bit full adder to simplify reuse in higher-level designs. The symbol includes three inputs (A, B, and C_{in}), two outputs (S and C_{out}), and one supply pin (V_{DD}). This symbol will be used in the next part to construct the 4-bit ripple-carry adder.



24) Full Adder Simulation

The one-bit NOR-based full adder was verified using transient simulation. Three pulse inputs were applied to A, B, and C_{in} with binary-weighted periods so that all eight possible input combinations were generated over an 80 ns interval. The simulated outputs S and C_{out} match the expected full-adder behavior. Specifically, the sum output follows the XOR relation of the three inputs, while the carry output is asserted whenever at least two inputs are high. The simulated waveforms confirm correct operation of the one-bit full adder.

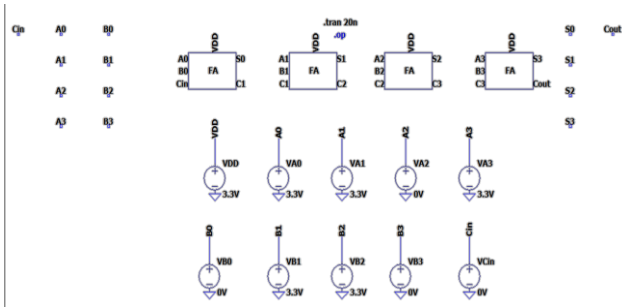




B. 4-Bit-Ripple-Carry-Adder

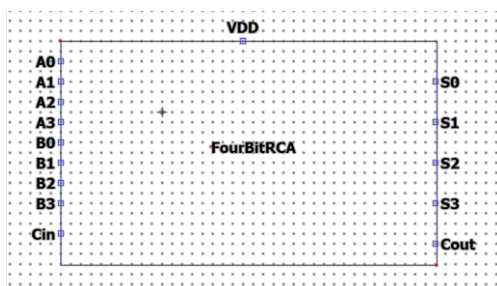
25) RCA Construction

A 4-bit ripple-carry adder was constructed by cascading four instances of the one-bit full-adder symbol created in Q23. The carry output of each stage was connected to the carry input of the next stage, forming a ripple-carry chain. The stage interconnections were implemented as follows: $FA_0(A_0, B_0, C_{in}) \rightarrow (S_0, C_1)$, $FA_1(A_1, B_1, C_1) \rightarrow (S_1, C_2)$, $FA_2(A_2, B_2, C_2) \rightarrow (S_2, C_3)$, and $FA_3(A_3, B_3, C_3) \rightarrow (S_3, C_{out})$. The complete 4-bit ripple-carry adder therefore has nine inputs: $A_3, A_2, A_1, A_0, B_3, B_2, B_1, B_0$, and C_{in} , and five outputs: S_3, S_2, S_1, S_0 , and C_{out} . All four stages were powered using the common supply voltage $V_{DD} = 3.3$ V.



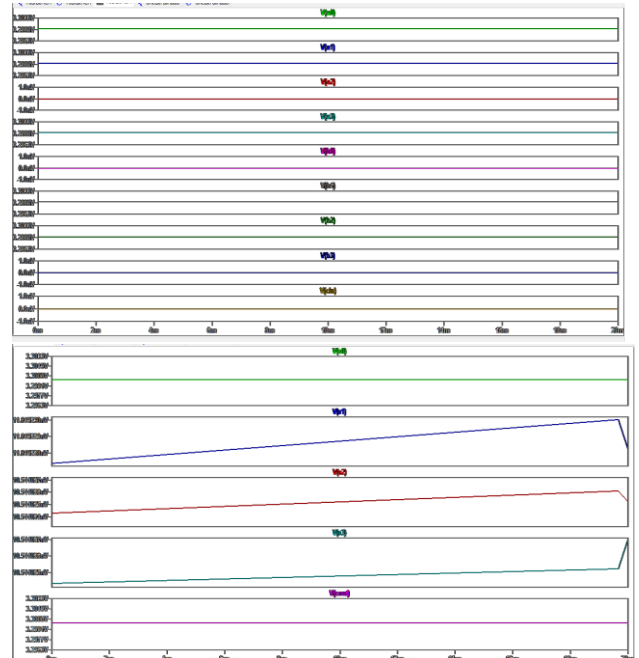
26) RCA Symbol

A hierarchical LTspice symbol was created for the 4-bit ripple-carry adder to simplify its use in higher-level simulations. The symbol includes the four-bit input buses $A_3:A_0$ and $B_3:B_0$, the carry input C_{in} , the four-bit sum outputs $S_3:S_0$, the carry output C_{out} , and the supply input V_{DD} . This hierarchical representation allows the complete 4-bit ripple-carry adder to be instantiated as a single reusable block, significantly reducing schematic complexity and improving design readability in larger digital circuits.



27) RCA Simulation

A 4-bit CMOS ripple-carry adder (RCA) was simulated in LTspice to verify correct binary addition and carry propagation through all four cascaded full-adder stages. A DC input combination was applied as follows: $A = 1011_2$, $B = 0110_2$, and $C_{in} = 0$. This corresponds to the decimal addition $11 + 6 + 0 = 17$, whose expected binary result is 10001_2 . Therefore, the expected outputs are $S_3S_2S_1S_0 = 0001$ and $C_{out} = 1$. A DC operating-point simulation was performed to validate the RCA operation. The simulated output voltages were $S_0 \approx 3.3$ V, $S_1 \approx 0$ V, $S_2 \approx 0$ V, $S_3 \approx 0$ V, and $C_{out} \approx 3.3$ V, which correspond to the logic output $C_{out}S_3S_2S_1S_0 = 10001$. The simulated result exactly matches the expected binary sum, confirming correct sum generation and carry propagation through all four full-adder stages. The simulation verifies that the 4-bit CMOS ripple-carry adder performs the intended binary addition correctly.



V. CONCLUSION

In this work, MOS and CMOS digital circuits were designed and simulated using the GF180MCU PDK in LTspice. The DN MOS RTL inverter and CMOS inverter were analyzed in both DC and transient operation, and the simulated results showed good agreement with hand calculations. A NOR-based full adder and 4-bit ripple-carry adder were then successfully implemented and verified. Overall, the results confirm correct circuit operation and demonstrate the practical behavior of digital CMOS design in the GF180MCU process.